Pizza Sales Analysis Report (SQL Project)

Project Overview

This project analyzes a pizza store dataset with **48,000+ records** to understand sales performance, customer ordering behavior, and product trends. The analysis is done using SQL by writing queries to calculate key business metrics and generate insights. The goal of this project is to support better business decisions using data.

Key Metrics

-- Total Revenue SELECT SUM(total_price) AS total_revenue FROM pizza_sales;	
-- Average Order Value SELECT ROUND(SUM(total_price) / COUNT(DISTINCT order_id), 2) AS average_order_price FROM pizza_sales;	
-- Total Pizzas Sold SELECT SUM(quantity) AS total_pizza_sold FROM pizza_sales;	
-- Total Orders SELECT COUNT(DISTINCT order_id) AS total_orders FROM pizza_sales;	
-- Average Pizzas per Order SELECT ROUND(SUM(quantity)::DECIMAL / COUNT(DISTINCT order_id), 2) AS avgerage_pizza_per_order FROM pizza_sales;	

Sales Trends

```
-- Daily Trend (Orders by Day)
SELECT
TO_CHAR(order_date, 'Day') AS order_day,
COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY order_day
ORDER BY total_orders DESC;
```

	order_day text	total_orders bigint
1	Friday	3538
2	Thursday	3239
3	Saturday	3158
4	Wednesday	3024
5	Tuesday	2973
6	Monday	2794
7	Sunday	2624

```
-- Hourly Trend (Orders by Hour)
SELECT
EXTRACT(HOUR FROM order_time) AS
order_hour,
COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY order_hour
ORDER BY total_orders DESC;
```

	order_hours numeric	total_orders bigint
1	9	1
2	10	8
3	23	28
4	22	663
5	21	1198
6	11	1231
7	15	1468
8	14	1472
9	20	1642
10	16	1920
11	19	2009
12	17	2336
13	18	2399
14	13	2455
15	12	2520

Category & Size Analysis

```
-- Sales by Category (January)
SELECT
pizza_category,
SUM(total_price) AS total_sales_category,
ROUND(
SUM(total_price) * 100.0 /
(SELECT SUM(total_price)
FROM pizza_sales
WHERE EXTRACT(MONTH FROM order_date) = 1),
2) AS percentage_of_sales
FROM pizza_sales
WHERE EXTRACT(MONTH FROM order_date) = 1
GROUP BY pizza_category;
```

	pizza_category character varying (50)	total_sales_category numeric	percentage_of_sales numeric
	Chicken	16188.75	23.1952780567762235
	Classic	18619.40	26.6779189406432996
	Supreme	17929.75	25.6897868420034588
	Veggie	17055.40	24.4370161605770181

```
-- Sales by Pizza Size
SELECT
pizza_size,
SUM(total_price) AS total_sales_category,
ROUND(
SUM(total_price) * 100.0 /
(SELECT SUM(total_price) FROM pizza_sales),
2) AS pct
FROM pizza_sales
GROUP BY pizza_size;
```

	pizza_size character varying (10)	total_sales_category numeric	pct numeric
1	S	178076.50	21.77
2	XXL	1006.60	0.12
3	XL	14076.00	1.72
4	M	249382.25	30.49
5	L	375318.70	45.89

Quantity Analysis

```
-- Total Pizzas Sold by Category
SELECT
pizza_category,
SUM(quantity) AS total_pizzas_sold
FROM pizza_sales
GROUP BY pizza_category;
```

	pizza_category character varying (50)	total_pizzas_sold bigint
1	Supreme	11987
2	Chicken	11050
3	Veggie	11649
4	Classic	14888

Top & Bottom Performers

```
-- Top 5 Best-Selling Pizzas
SELECT
pizza_name,
SUM(quantity) AS total_pizzas_sold
FROM pizza_sales
GROUP BY pizza_name
ORDER BY total_pizzas_sold DESC
LIMIT 5;
```

	pizza_name character varying (100)	total_pizzas_sold bigint
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Piz...	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

```
-- Bottom 5 Worst-Selling Pizzas
SELECT
pizza_name,
SUM(quantity) AS total_pizzas_sold
FROM pizza_sales
GROUP BY pizza_name
ORDER BY total_pizzas_sold ASC
LIMIT 5;
```

	pizza_name character varying (100)	total_pizzas_sold bigint
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Piz...	950
5	The Soppressata Pizza	961

Key Business Insights

- Most revenue comes from a few top-performing categories → focus marketing here
- Peak sales occur during specific hours → optimize staffing and delivery
- Some pizzas have very low demand → possible menu optimization
- Average order size shows opportunity for combo deals or upselling
- Larger size pizzas may contribute more revenue → pricing strategy can focus here

Conclusion

This SQL analysis helps understand customer behavior, product performance, and sales trends. The insights can be used to improve marketing strategies, optimize menu offerings, and increase overall business revenue. This project demonstrates strong skills in SQL, data analysis, and business insight generation.